

# **Effective Programming Practices for Economists**

## **Python Installation and Execution**

### **Modifying Pixi environments**

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# The pyproject.toml file

```
# pyproject.toml

[tool.pixi.project]
name = "toy-project"
channels = ["conda-forge"]
platforms = [
    "linux-64",
    "osx-64",
    "osx-arm64",
    "win-64",
]

[tool.pixi.dependencies]
python = ">=3.13"
numpy = ">=2.2.3"
```

- The `pyproject.toml` file can be used to configure pixi for the project
- Dependencies are tracked in the `[tool.pixi.dependencies]` table
- Will see two ways of changing this:
  - Using `pixi add [package]` in a shell
  - Modifying it directly in `pyproject.toml`

# What is in the environment?

```
$ pixi list
```

| Package          | Version   | Build                | Size      | Kind  | Source           |
|------------------|-----------|----------------------|-----------|-------|------------------|
| _libgcc_mutex    | 0.1       | conda_forge          | 2.5 KiB   | conda | _libgcc_mutex    |
| _openmp_mutex    | 4.5       | 2_gnu                | 23.1 KiB  | conda | _openmp_mutex    |
| bzip2            | 1.0.8     | h4bc722e_7           | 246.9 KiB | conda | bzip2            |
| ca-certificates  | 2025.1.31 | hbcca054_0           | 154.4 KiB | conda | ca-certificates  |
| ld_impl_linux-64 | 2.43      | h712a8e2_4           | 655.5 KiB | conda | ld_impl_linux-64 |
| libblas          | 3.9.0     | 31_h59b9bed_openblas | 16.5 KiB  | conda | libblas          |
| libcblas         | 3.9.0     | 31_he106b2a_openblas | 16.4 KiB  | conda | libcblas         |
| ...              |           |                      |           |       |                  |
| libstdcxx        | 14.2.0    | h8f9b012_2           | 3.7 MiB   | conda | libstdcxx        |
| libuuid          | 2.38.1    | h0b41bf4_0           | 32.8 KiB  | conda | libuuid          |
| libzlib          | 1.3.1     | hb9d3cd8_2           | 59.5 KiB  | conda | libzlib          |
| ncurses          | 6.5       | h2d0b736_3           | 870.7 KiB | conda | ncurses          |
| numpy            | 2.2.3     | py313h17eae1a_0      | 8.1 MiB   | conda | numpy            |
| openssl          | 3.4.1     | h7b32b05_0           | 2.8 MiB   | conda | openssl          |
| python           | 3.13.2    | hf636f53_101_cp313   | 31.7 MiB  | conda | python           |
| python_abi       | 3.13      | 5_cp313              | 6.1 KiB   | conda | python_abi       |
| readline         | 8.2       | h8c095d6_2           | 275.9 KiB | conda | readline         |
| tk               | 8.6.13    | noxft_h4845f30_101   | 3.2 MiB   | conda | tk               |
| tzdata           | 2025a     | h78e105d_0           | 120 KiB   | conda | tzdata           |

# Adding dependencies

```
$ pixi add pandas
```

```
# pyproject.toml
```

```
[tool.pixi.project]
name = "toy-project"
channels = ["conda-forge"]
platforms = [
    "linux-64",
    "osx-64",
    "osx-arm64",
    "win-64",
]
```

```
[tool.pixi.dependencies]
python = ">=3.13"
numpy = ">=2.2.3"
pandas = ">=2.2.3, <3"
```

# Adding dependencies (2)

- You can change the `pyproject.toml` file directly:
  - Add dependencies to the `[tool.pixi.dependencies]` table
  - When calling `pixi run` or `pixi install`, pixi will install the dependencies
- **Version constraints**
  - pandas version 2.2 or higher: `pandas = ">=2.2"`
  - pandas version 2.2.x: `pandas = "~=2.2"`
  - pandas version 2.2.0: `pandas = "==2.2.0"`
  - most recent version (*that is compatible with other packages*): `pandas = "*"`

# Adding PyPi dependencies

```
$ pixi add --pypi pdbp
```

```
# pyproject.toml
```

```
[tool.pixi.project]
name = "toy-project"
channels = ["conda-forge"]
platforms = [
    "linux-64",
    "osx-64",
    "osx-arm64",
    "win-64",
]

[tool.pixi.dependencies]
python = ">=3.13"
numpy = ">=2.2.3"
pandas = ">=2.2.3, <3"

[tool.pixi.pypi-dependencies]
pdbp = "*"

```

# The lock file

- Upon installation of dependencies, pixi creates the `pixi.lock` file
- The lock file contains the exact versions of the dependencies that were installed
- The lock file is automatically updated when any dependency is changed
- This ensures reproducibility of the environment / results
- The lock file **should** be committed to the repository